Glomerular Disease — Nephrotic Syndrome

Pathology GUE Block · MEDucation by Mattin Majid

Continued in Part II: this LG covers two syndromes — nephrotic syndrome (here) and nephritic syndrome (Part II).

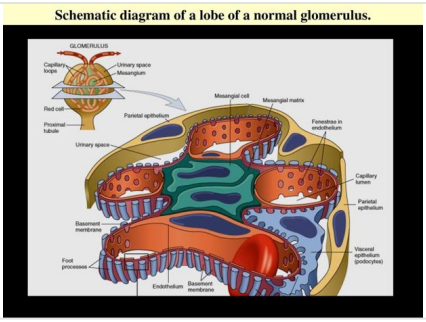
BEFORE YOU READ — ANSWER FROM MEMORY

1. What is the key difference in proteinuria selectivity between minimal change disease and membranous glomerulopathy, and what does each imply about corticosteroid response?
2. What EM finding distinguishes membranous GN (subepithelial deposits/spikes) from MPGN (GBM duplication/tram-tracking) from diabetic nephropathy (nodular Kimmelstiel-Wilson lesions)?
3. Which WHO lupus nephritis class shows the wire-loop lesion, and what deposit location produces it?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The glomerulus & glomerular disease — the framework

What	The glomerulus is a tuft of capillaries within a Bowman capsule at the end of a renal tubule; it is the basic filtration unit of the kidney.
GN definition	Glomerulonephritis (GN) / Bright disease = diseases that primarily involve the renal glomeruli. Two broad groups: **(I) Primary GN** — glomeruli are the predominant site of involvement; **(II) Secondary glomerular diseases** — certain systemic and hereditary diseases that secondarily affect the glomeruli.
Histologic terms	Classification is based on histology: **diffuse** (all glomeruli), **global** (entire glomerulus), **focal** (portion of glomeruli), **segmental** (part of each glomerulus), **mesangial** (affecting mesangial region).
Pathogenesis	(1) **Circulating immune complex deposition** — antigen + antibody deposit in glomeruli, bind complement, cause inflammation and glomerular injury (antigen may be exogenous or endogenous). (2) **Antibodies directed against antigen on the glomerular capillary membrane** — anti-GBM antibody disease (antigen fixed in the GBM, e.g. Goodpasture syndrome) or antibodies against a non-GBM antigen.



Schematic diagram of a lobe of a normal glomerulus

2 Nephrotic syndrome — definition & causes

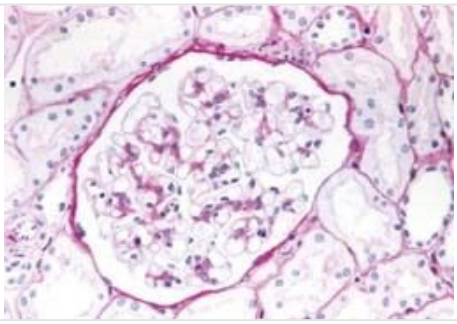
What	A clinical complex characterized by **heavy proteinuria (≥3.5 g/day)**, **hypoalbuminemia** (plasma protein <3 g), **edema**, **hyperlipidemia**, and **hyperlipiduria** (increased hepatic lipoprotein synthesis).
Proteinuria mechanism	Injury to the glomerular capillary wall increases its permeability to plasma protein, allowing it to escape from plasma (daily loss ≥3.5 g).

Category	Causes
I. Primary GN	Minimal change disease (most common in **children**); membranous GN (most common in **adults**); membranoproliferative GN; focal segmental glomerulosclerosis; focal and diffuse proliferative GN; IgA nephropathy.

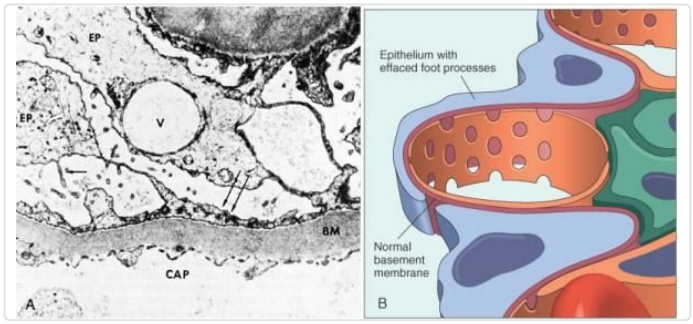
II. Systemic diseases	Diabetes mellitus; amyloidosis; SLE.
III. Systemic infections	Viral (HBV, HCV, HIV); bacterial (bacterial endocarditis, syphilis, leprosy); protozoa and parasites (*P. falciparum* malaria, filariasis).
IV. Hypersensitivity reactions	Drugs (heavy metals — gold, mercury; penicillamine, trimethadione, tolbutamide; heroin addiction); bee stings, snake bite, poison ivy.
V. Malignancy	Carcinomas; myeloma; Hodgkin disease.
VI. Pregnancy	Toxemia of pregnancy.
VII. Circulatory disturbances	Renal vein thrombosis; constrictive pericarditis.
VIII. Hereditary diseases	Alport disease; Fabry disease; nail-patella syndrome.

3 Minimal change disease (Lipoid Nephrosis)

WHO	**Most frequent cause of NS in children**; peak incidence 2–6 years. Follows a respiratory infection or prophylactic immunization.
Proteinuria	**Highly selective** (albumin).
Treatment response	**Dramatic response to corticosteroid therapy.**
Mechanism	Tubular reabsorption of lipoproteins passing through diseased glomeruli (hence "lipoid nephrosis").
Morphology	Light microscopy: glomeruli are **normal**. Immunofluorescence: **no immunoglobulin or complement deposits.** Electron microscopy: GBM appears normal, no electron-dense material deposited; visceral epithelial cells show **diffuse effacement of foot processes**, replaced by a rim of cytoplasm showing vacuolization and hyperplasia of villi.



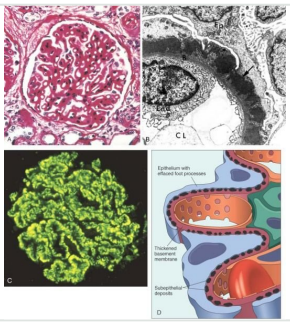
MCD — PAS stain: normal basement membrane, absence of proliferation



EM — effacement of foot processes (double arrows), absence of deposits, vacuoles (V), microvilli

4 Membranous glomerulopathy (MG)

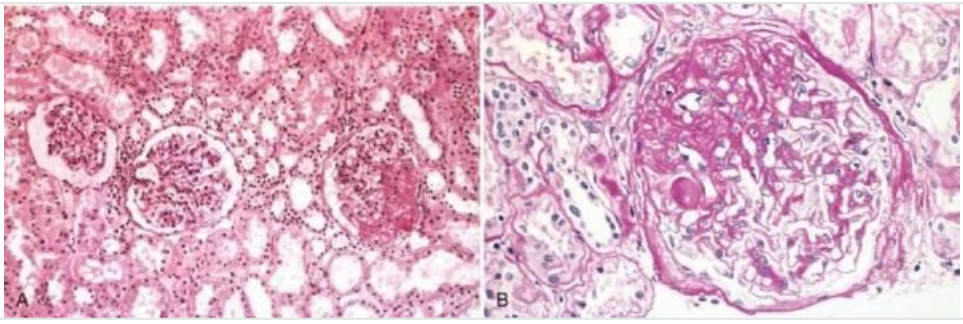
WHO	**Most common cause of NS in adults.**
Cause	**Primary MG** (85% of cases) — idiopathic (renal auto-antigen). **Secondary MG** — drugs (penicillamine); malignancy (Ca of lung, colon, melanoma); SLE; infections (chronic hepatitis B or C, malaria, syphilis); autoimmune thyroiditis.
Pathogenesis	**Immune complex-mediated disease.** In contrast to minimal change disease, proteinuria is **nonselective** (albumin and globulin both excreted). Does not respond well to corticosteroid therapy.
Morphology	Light microscopy: diffuse thickening of glomerular basement membrane. Electron microscopy: apparent thickening of GBM caused by **subepithelial deposits** separated from each other by small spikes in the GBM matrix.



A: PAS — diffuse capillary wall thickening without increased cellularity. B: EM — electron-dense subepithelial deposits with obliteration of foot processes. C: granular immunofluorescent IgG deposits along GBM. D: diagrammatic representation

5 Focal Segmental Glomerulosclerosis (FSGS)

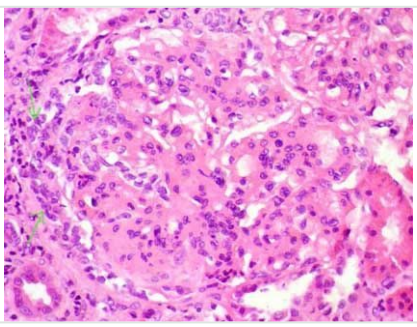
What	Sclerosis of **some** glomeruli (focal), and in the affected glomeruli only a portion is involved (segmental). Sclerosis occurs within capillary tufts of deep juxtamedullary glomeruli.
Clinically	Nephrotic syndrome.
Causes	Genetic; drugs (anabolic steroids), toxins (heroin); infection (HIV); chronic pyelonephritis, diabetes, and hypertension; idiopathic.



FSGS, PAS stain. A: low power — segmental sclerosis in one of three glomeruli (3 o'clock). B: high power — hyaline insudation and lipid (small vacuoles) in the sclerotic area

6 Membranoproliferative glomerulonephritis (MPGN)

Current classification (by immunofluorescence)	Immunoglobulin-mediated (monoclonal or polyclonal) or complement-mediated — immune complex/monoclonal immunoglobulin mediated MPGN (activation of classic complement pathway), or C3 glomerulopathies — presenting with protein- or hematuria.
Primary vs secondary	Primary: idiopathic. Secondary MPGN arises in SLE, hepatitis B or C infection, α1-antitrypsin deficiency, and malignant diseases (CLL, lymphoma). *Previous classification* subdivided primary MPGN into Type I and Type II (dense-deposit disease) on ultrastructural/immunofluorescent/ pathologic grounds.
Morphology (light microscopy)	Glomeruli are large, lobular, and **hypercellular** — due to proliferation of cells in the mesangium and endocapillary cell proliferation (capillary endothelium and infiltrating leukocytes). Increased mesangial matrix. GBM thickened, often focally. Capillary wall shows a "**double-contour**" or "**tram-track**" appearance caused by duplication of the GBM.



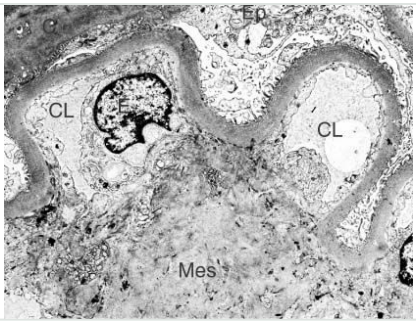
MPGN — endocapillary and mesangial hypercellularity (H&E)



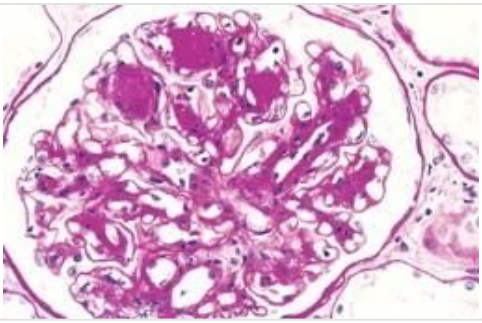
Duplication of GBM with double-contour / tram-track appearance (silver stain)

7 Diabetic nephropathy

Clinically	Three glomerular syndromes: (1) non-nephrotic proteinuria, preceded by microalbuminuria; (2) nephrotic syndrome; (3) chronic renal failure. Diabetes also causes hyalinizing arteriolar sclerosis, and increased susceptibility to acute pyelonephritis and papillary necrosis.
Pathogenesis	Two processes: **metabolic defect** — advanced glycosylation end products lead to thickened GBM and increased mesangial matrix; **hemodynamic effects** — associated with glomerular hypertrophy which contributes to glomerulosclerosis.
Morphology	GBM thickening. Increase in mesangial matrix results in: **diffuse glomerulosclerosis**, or **nodular glomerulosclerosis (Kimmelstiel-Wilson disease)** — ovoid or spherical laminated PAS-positive nodules lie within the mesangial core, surrounded by dilated capillary loops.



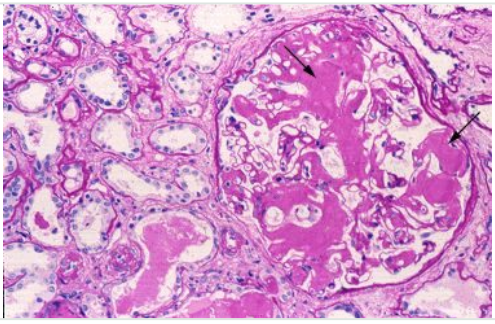
EM — massive increase in mesangial matrix (Mes) encroaching on capillary lumen (CL); GBM and Bowman capsule markedly thickened



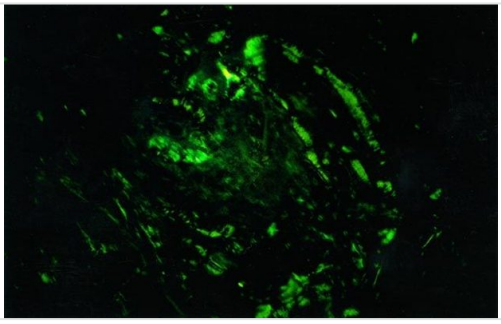
Nodular glomerulosclerosis (Kimmelstiel-Wilson), PAS stain

8 Renal amyloidosis

What	Another cause of nephrotic syndrome. Subendothelial and mesangial amyloid deposits.
Identification	Congo red stain gives: **red color under light microscopy**; **apple-green birefringence under polarized light**. Cross-fibrillary pattern of amyloid by electron microscopy.
Two types	**Primary (AL) amyloidosis** — in patients with plasma cell dyscrasias (multiple myeloma); consists of monoclonal light chains (kappa or lambda). **Secondary (AA) amyloidosis** — in patients with neoplasia or chronic inflammation (e.g. TB); associated with serum amyloid A protein produced in the liver.



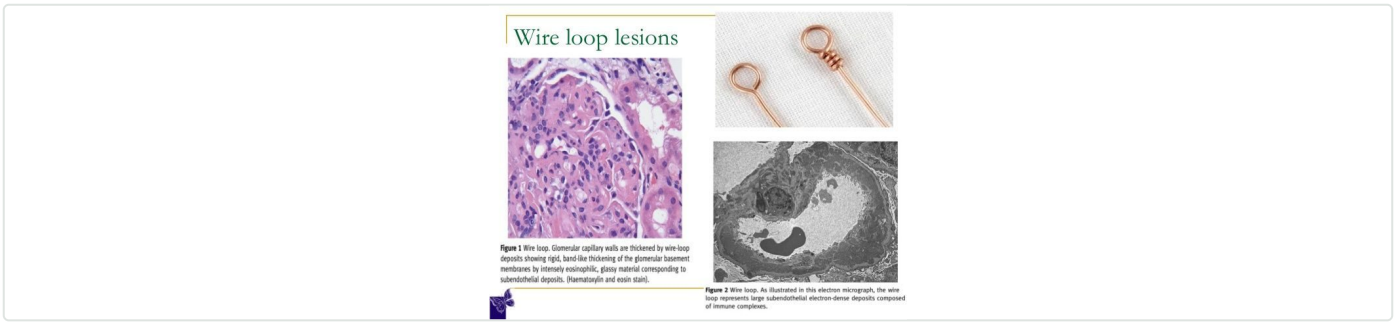
Amyloid deposits (PAS stain, arrows)



Amyloid — immunofluorescence

9 Lupus nephropathy

WHO	The severity of renal lesion often determines the prognosis. Classified (WHO) into **six renal patterns**: **Type I** — no observable renal involvement. **Type II** — mesangial proliferative form. **Type III** — focal proliferative form. **Type IV** — diffuse proliferative form; characteristic change includes **wire-loop lesion** (wire-loop thickening of GBM from subendothelial immune complex deposition). **Type V** — membranous form (global or segmental subepithelial immune deposits). **Type VI** — advanced sclerosis lupus nephritis.



Wire loop lesion — rigid, band-like, eosinophilic thickening of GBM (subendothelial deposits); EM shows large subendothelial electron-dense immune-complex deposits

P Must-remember pearls

- MCD: normal glomeruli on LM, no deposits on IF, **foot process effacement only** on EM; **selective** proteinuria; dramatic steroid response. Most common NS cause in **children**.
- Membranous GN: **subepithelial deposits with GBM spikes**; **nonselective** proteinuria; poor steroid response. Most common NS cause in **adults**; 85% idiopathic.
- FSGS: sclerosis of **some** glomeruli, **part** of each — classically affects deep juxtamedullary glomeruli first.
- MPGN: hypercellular, lobular glomeruli with **"tram-track"/double-contour** GBM from duplication — due to mesangial interposition.
- Diabetic nephropathy: metabolic (AGEs → GBM thickening) + hemodynamic (glomerular hypertrophy) mechanisms; classic lesion is **Kimmelstiel-Wilson nodular glomerulosclerosis.**
- Amyloid: **Congo red** — red under light microscopy, **apple-green birefringence** under polarized light. AL = light-chain (myeloma); AA = serum amyloid A (chronic inflammation).
- Lupus nephritis **Class IV** (diffuse proliferative) is the most severe and shows the classic **wire-loop lesion** from subendothelial immune complex deposition.

Q Test yourself

1. A 4-year-old develops nephrotic syndrome after a URI. Renal biopsy shows normal glomeruli on light microscopy, no immune deposits on IF, and diffuse foot process effacement on EM. What is the diagnosis?

A. Membranous glomerulopathy
B. Minimal change disease
C. FSGS
D. MPGN

2. Which glomerular disease classically shows nonselective proteinuria, subepithelial deposits, and GBM spikes on EM, and responds poorly to steroids?

A. Minimal change disease
B. Membranous glomerulopathy
C. Diabetic nephropathy
D. IgA nephropathy

3. A diabetic patient's renal biopsy shows ovoid, laminated PAS-positive nodules within the mesangial core surrounded by dilated capillary loops. This finding is called:

A. Wire-loop lesion
B. Kimmelstiel-Wilson nodules
C. Tram-track appearance
D. Crescent formation

4. A renal biopsy stained with Congo red shows apple-green birefringence under polarized light. The patient has multiple myeloma. Which type of amyloid is most likely?

A. AA amyloid
B. AL amyloid
C. Transthyretin amyloid
D. Beta-2 microglobulin amyloid

5. A lupus patient's renal biopsy shows diffuse proliferative glomerulonephritis with wire-loop lesions from subendothelial immune complex deposition. Which WHO class is this?

A. Class II
B. Class III
C. Class IV
D. Class V

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — Normal glomeruli on LM, no IF deposits, and isolated foot process effacement on EM define minimal change disease, the most common cause of childhood NS.

2 · B — Membranous glomerulopathy shows nonselective proteinuria, subepithelial deposits separated by GBM spikes, and poor steroid response, unlike MCD.

3 · B — Ovoid/spherical laminated PAS-positive nodules in the mesangial core define Kimmelstiel-Wilson nodular glomerulosclerosis of diabetic nephropathy.

4 · B — Multiple myeloma (plasma cell dyscrasia) produces AL amyloid, composed of monoclonal light chains.

5 · C — Diffuse proliferative lupus nephritis with wire-loop lesions from subendothelial deposits is WHO Class IV, the most severe class.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam